

# Turán number of a finite family.

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| OPG-46706 | L1 - Tractable | Partiellement résolu | Graph Theory |
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## Énoncé original (anglais, reproduction mot pour mot)

Given a finite family  $\{\mathcal{F}\}$  of graphs and an integer  $n$ , the Turán number  $ex(n, \{\mathcal{F}\})$  of  $\{\mathcal{F}\}$  is the largest integer  $m$  such that there exists a graph on  $n$  vertices with  $m$  edges which contains no member of  $\{\mathcal{F}\}$  as a subgraph.

Conjecture For every finite family  $\{\mathcal{F}\}$  of graphs there exists an  $F \in \{\mathcal{F}\}$  such that  $ex(n, \mathcal{F}) = O(ex(n, F))$ .

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